



NTU Academy for Professional
and Continuing Education

(SCTP) Advanced
Professional Certificate

Data Science and AI



Coding Class Norms

For a Smooth & Productive Session

- **Maintaining Class Flow**
 - Complete all pre-work — class builds on it.
 - If pre-work isn't done: listen, observe, catch up later.
 - If majority face the same issue, we will pause to address together.
- **Self Sufficiency**
 - Review instructions/materials.
 - Search online — Official docs, Google Search, Stack Overflow, AI assistants.
 - Attempt troubleshooting
- **Peer Learning**
 - Build relationship, collaborate with your buddy or assigned group to solve problems.
 - Supporting peers strengthens teamwork, deepens understanding,
 - Enhance learning from diverse skills and viewpoints



NANYANG
TECHNOLOGICAL
UNIVERSITY
SINGAPORE

1.3 SQL Basics - DDL

Module Overview

1.1 Introduction to Data Science

1.2 Introduction to Database

1.3 SQL Basic - DDL

1.4 SQL Basic - DML

1.5 SQL Advanced

1.6 Introduction to Numpy

1.7 Introduction to Pandas

1.8 Exploratory Data Analysis (EDA) Basic

1.9 EDA Advanced

1.10 Data Visualisation

Recap Last Lesson

- In your own words, describe the following:
 - Database History
 - Database types and use cases
 - ERD Construction & Keys (PK/FK)
 - Normalization (1NF-3NF)

Lesson Objectives

- SQL Data Definition Language (DDL) statements
- Databases, schemas, tables, keys, indexes, constraints
- Tables (materialized) vs Views (virtual)

SQL Data Definition Language



<https://www.youtube.com/watch?v=adACCuMFjac>

Hands on Exercise with DbGate

1. Connecting to database
2. Executing SQL DDL statements
3. Executing SQL DDL statements based on ERD
4. Tables vs Views
5. Importing / Exporting Data

Class Activity

<https://github.com/su-ntu-ctp/6m-data-1.3-sql-basic-ddl/blob/main/lesson.md>